



Tecnológico de Monterrey  
Interpartial Exam 3 A  
2026-Jan-May

Grade

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Course: Calculus I

ID-number: \_\_\_\_\_

Professor: Fco. Javier Vazquez Tavares

Date: March 9, 2026

Last Names and Names: \_\_\_\_\_

Group: 201

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*"In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honesty and not to use unauthorized or illegal means to do so."*

Signature: \_\_\_\_\_

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**General instructions.** Solve the derivatives and reduce the final answer to the greatest extent possible using the information given.

$$\begin{aligned}\frac{d}{dx}u(x)v(x) &= u'(x)v(x) + u(x)v'(x) \\ \frac{d}{dx}\frac{u(x)}{v(x)} &= \frac{1}{v^2(x)}[u'(x)v(x) - u(x)v'(x)] \\ \frac{d}{dx}u(v(x)) &= u'(v(x))v'(x)\end{aligned}$$

$$\begin{aligned}a^n \cdot a^m &= a^{n+m}, & \log_n[a] + \log_n[b] &= \log_n[a \cdot b] \\ (a^n)^m &= a^{n \cdot m}, & a \log_n[b] &= \log_n[b^a] \\ \frac{1}{a} &= a^{-1}, & \log_n[a] - \log_n[b] &= \log_n\left[\frac{a}{b}\right] \\ a^{\log_a[x]} &= x = \log_a[a^x], & \log_n[x] &= \frac{\log_m[x]}{\log_m[n]}\end{aligned}$$

**I** Product, Quotient and Chain Rule. Solve the following derivatives using the product, quotient and chain rule.

**1.** (16 points) Compute the derivative of  $f(x) = e^{\sin(x)}$ .

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2. (16 **points**) Compute the derivative of  $f(x) = \frac{2}{1 + e^x}$ .

3. (16 **points**) Compute the derivative of  $f(x) = e^{-x^2} \sin(x)$

**II** Derivative of exponential and logarithmic functions. Compute the derivative of the following functions. You can use the exponential and logarithmic rules or use the chain rule.

1. (16 **points**) Compute the derivative of the function  $f(x) = \log_2 [2^x + 1]$

2. (16 **points**) Compute the derivative of the function  $f(x) = \left(\frac{1}{4}\right)^x$

### III Critical Points.

1. (20 **points**) Find the critical point of the following function  $f(x) = 6x \cdot e^{3/2x}$ . 10 points are assign to the derivative and the other 10 points are assign to the critical point.